Automated Irregular Shaped Stone Stacking via Pose Optimization and Monte Carlo Tree Search

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***Abstract*—We study automated stacking of irregular objects to approximate prescribed target shapes without support from container walls. Our framework combines bilevel pose optimization with batched, receding-horizon Monte Carlo Tree Search (MCTS) over object choices and placement poses. The optimizer couples geometric objectives with contact-force optimization to evaluate force and torque equilibrium across the structure under friction constraints, using a bodywise decomposition for GPU parallelization. Physical simulation validates placements and supplies settled successor states for multi-step search. We hypothesize that combining structural stability evaluation with lookahead improves target coverage while maintaining stability under a fixed planning budget.**



Fig. 1: Preliminary stacking results. Left: simulated structures for different target shapes. Right: a physical stone structure from an outdoor experiment.

I. Introduction

Robotic packing methods accommodate irregular objects while considering packing density, stability, and placement feasibility [22, 9]. Constructing freestanding structures from natural stones requires finding mutually supporting arrangements among uneven surfaces while achieving a prescribed target shape without support from container walls.

Several stone-stacking methods use placement criteria developed for particular construction settings, including tower-building objectives [7] and masonry-inspired geometric rules [12, 19]. For example, Johns et al. generate candidates where target wall surfaces meet the existing structure and apply inward-slope heuristics [12]. Extensions to freeform walls [13] broaden wall geometry; more general targets motivate objectives that use target geometry without wall-specific placement rules. Structural stability also requires reasoning beyond the support of the newly placed object: its load can affect the equilibrium of objects beneath it. Liu and Napp [17] explicitly address structural strength during planning, demonstrating the value of evaluating the structure as a whole. However, assessing the current structure alone does not determine how a placement affects future support and space utilization. Menezes et al. [20] address sequential placement through learned action values that account for future rewards, with a prescribed object sequence and fixed orientations. We investigate explicit lookahead over object choices and placement poses, coupled with structural stability evaluation and target-shape objectives.

We consider stacking a finite inventory of n irregular objects to approximate a target shape $T \subset \mathbb{R}^3$. Let $O_i(q_i) \subset \mathbb{R}^3$ denote the volume occupied by object i at pose $q_i \in \mathrm{SE}(3)$. The final configuration objective is

$$\min_{q_1,\cdots,q_n \in \mathrm{SE}(3)} \mathbb{V}\left(T \setminus \bigcup_{i=1}^{n} O_i(q_i)\right) \quad \text{s.t.} \quad c_{\text{phys}}(O_1(q_1),\cdots,O_n(q_n)) = 0, \tag{1}$$

where $\mathbb{V}(\cdot)$ denotes volume and $c_{\text{phys}} = 0$ represents stability and non-interpenetration constraints. Jointly optimizing all object poses under coupled physical constraints is computationally demanding. We therefore select one object and optimize its placement at each step, checking the resulting structure's physical feasibility. Each placement changes support geometry, free space, and remaining inventory, motivating lookahead beyond immediate placement quality. Our central question is how to combine this lookahead with structural stability evaluation in a common framework for different target shapes.

Our framework couples bilevel pose optimization with batched, receding-horizon Monte Carlo Tree Search (MCTS) [8]. The optimizer combines target geometry with structure-wide force and torque equilibrium under friction constraints, using bodywise ADMM to parallelize contact-force optimization. Different targets share this formulation through their geometric representations. MCTS compares placement sequences using simulation to reject unstable placements and supply settled successor states.

We hypothesize that combining structural stability evaluation with multi-step lookahead improves target coverage while maintaining stability across different target geometries under a fixed planning budget. Comparisons with greedy placement and candidate-only stability evaluation will test the effects of

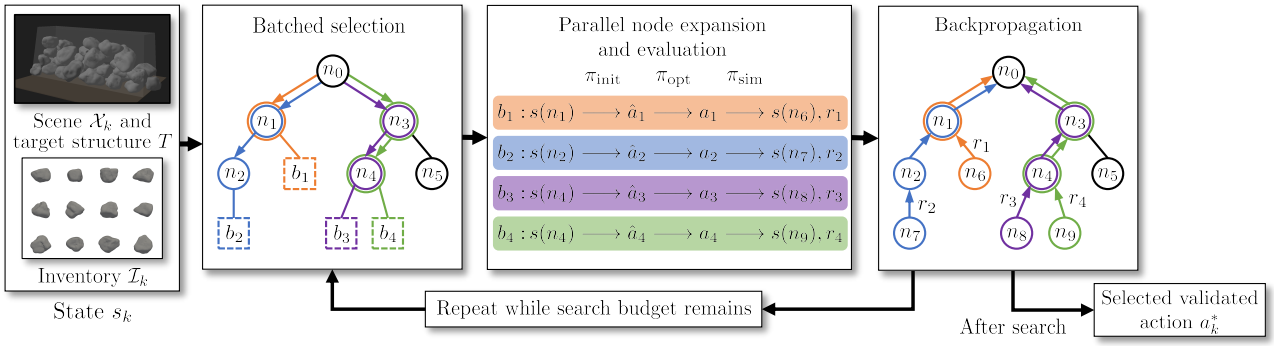


Fig. 2: Batched MCTS sequence planning. Colors identify expansion requests b_j and their tree paths. Requests are evaluated in parallel through π_{init}, π_{opt}, and π_{sim}, and returns are backpropagated along the selected paths. Search repeats within the budget before selecting a physically feasible root action a_k^* from the evaluated tree.

lookahead and structural coupling, respectively. Evaluations across target shapes and search horizons will examine the applicability of the common formulation and its computational tradeoffs. The proposed contributions are:

- A bilevel placement optimizer that combines explicit target geometry with contact-force optimization across the structure, using bodywise ADMM updates for GPU computation.
- A batched, receding-horizon stacking planner that integrates pose refinement and physical simulation with multi-step lookahead over object choices and placement poses.

II. Related Works

A. Automated Stone Stacking

Furrer et al. [7] combined gradient-based pose search with physics simulation for stone stacking. Johns et al. [12] integrated perception, planning, and excavator manipulation for wall construction, later using constrained registration and signed-distance-field classification for prescribed wall geometries [13]. Liu et al. [19] selected simulation-generated poses using hierarchical filters informed by masonry heuristics. Liu and Napp [17] incorporated structural strength evaluation into sequence planning with physical validation, later introducing stackability measures for corbelled shelter construction [18]. Menezes et al. [20] used deep Q-learning on elevation maps to select planar placements, rewarding target occupancy and penalizing object displacement. Their fixed-sequence, fixed-orientation construction strategies inform our initial pose sampler.

B. Irregular Object Packing

Wang and Hauser [22] packed nonconvex objects through heightmap minimization, considering stability, robot motion feasibility, and replanning for execution errors. Huang et al. [9] jointly planned object order and placement through hierarchical reinforcement learning with multi-view heightmaps and compactness, pyramidality, and simulated-stability rewards.

C. Contact-Aware Pose Optimization

Lee et al. [16] use differentiable Contact Factor Graphs for gradient-based manipulation planning. BODex [5] couples lower-level contact-force optimization with upper-level grasp synthesis, while GraspQP [23] defines a differentiable force-closure energy through a quadratic program. Our optimizer couples candidate placement with structure-wide equilibrium, using bodywise force decomposition for parallel evaluation.

D. Simulation-Based Sequential Manipulation Planning

King et al. [14] extended MCTS to nonprehensile rearrangement under uncertainty using learned and heuristic rollout policies. Huang et al. [8] introduced Parallel Monte Carlo Tree Search with Batched Simulations (PMBS), coupling parallel search with GPU rigid-body simulation for object retrieval from clutter. Our stacking planner adopts batched evaluation, combining geometric sampling, stability-aware pose refinement, and structural simulation at each expansion.

III. Overview and Preliminaries

A. Shape Representation

We represent the target T and each object O_i as unions of smooth convex bodies modeled by differentiable support functions (DSFs) [15]. For a compact convex body $C \subset \mathbb{R}^3$, its support function is $h_C(x) = \max_{p \in C} x^\top p$ for $x \neq 0$. At pose $q = (R, t) \in \mathrm{SE}(3)$, we write

$$C(q) = \{Rp + t \mid p \in C\}, \qquad O_i(q_i) = \bigcup_j C_{ij}(q_i),$$

where C_{ij} are the convex components in object i's local frame. Similarly, $T = \bigcup_j T_j$, where T_j are fixed convex components in the world frame. DSFs support differentiable signed-distance and contact-geometry computations between posed convex bodies [2], enabling gradient-based pose optimization.

B. Markov Decision Process Formulation

To address the construction objective in Eq. (1), we model stacking as a finite-horizon sequential decision process in which one object is placed at each step. Let $\mathcal{X}_k = \{(i, q_i) \mid$

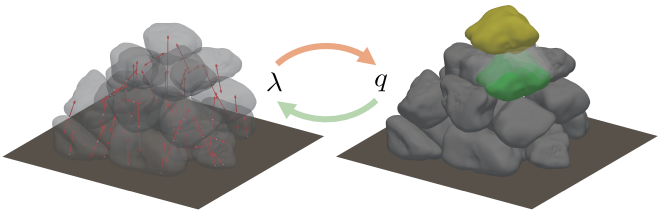


Fig. 3: Bilevel pose optimization. Left: optimizing contact forces λ under friction constraints minimizes structural equilibrium residuals at fixed q. Right: refining $q \in \mathrm{SE}(3)$ minimizes the geometric and optimized physical objectives.

$i \in \mathcal{P}_k\}$ be the current scene and $\mathcal{I}_k = \{1, \ldots, n\} \setminus \mathcal{P}_k$ the remaining inventory. The state, action, and deterministic transition function are

$$s_k = (\mathcal{X}_k, \mathcal{I}_k, T), \quad a_k = (i_{k+1}, q_{k+1}), \quad s_{k+1} = F(s_k, a_k),$$

where $i_{k+1} \in \mathcal{I}_k$. The transition inserts the proposed pose into the scene, simulates the resulting structure, and returns the settled scene with $\mathcal{I}_{k+1} = \mathcal{I}_k \setminus \{i_{k+1}\}$. The reward $r(s_k, a_k)$ scores target-contained volume and physical quality; feasibility checks separately reject invalid transitions (Sec. V). The sequence planner therefore operates on the MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, F, r, \gamma)$.

C. Monte Carlo Tree Search

Monte Carlo Tree Search (MCTS) incrementally constructs a search tree $\mathcal{T} = (\mathcal{V}, \mathcal{E})$ over the MDP. Each node $n \in \mathcal{V}$ represents a state $s(n)$, and each edge $(n, a) \in \mathcal{E}$ represents an action and its transition to a child node. Let $N(n)$ denote the number of completed evaluations and $Q(n)$ the empirical mean of discounted returns backpropagated to that node.

Each iteration balances value and exploration during *selection*, then generates and evaluates a new action during *expansion* and *evaluation*. *Backpropagation* updates $Q(n)$ and $N(n)$ along the selected path using the discounted return.

Our geometry-informed proposals, batched selection, and finite-horizon leaf evaluation are described in Sec. V.

D. Overall Structure of the Sequence Planner

Figure 2 illustrates the MCTS sequence planner. For each expansion, the initial pose sampler π_{init} proposes a candidate action $\hat{a}_k = (i_{k+1}, \hat{q}_{k+1})$, specifying both the object index and its initial pose. The pose optimization module π_{opt} refines the pose to obtain $a_k = (i_{k+1}, q_{k+1})$. The simulation module π_{sim} evaluates the placement and its effect on the structure, producing the settled successor state before disturbance trials. Validity and simulation statistics determine tree admission and reward, giving the execution flow

$$s_k \xrightarrow{\pi_{\text{init}}} \hat{a}_k \xrightarrow{\pi_{\text{opt}}} a_k \xrightarrow{\pi_{\text{sim}}} s_{k+1}, \; r(s_k, a_k).$$

IV. Pose Optimization

The module π_{opt} refines $\hat{q}_{k+1}$ from π_{init} toward a stable, physically feasible placement q_{k+1}.

A. Single Object Pose Optimization Problem

Given s_k, we initialize q at $\hat{q}_{k+1}$ and solve the bilevel problem

$$\begin{aligned} \min_{q} \quad & \left[f_{\text{geo}}(q) + \min_{\lambda} f_{\text{phys}}(q, \lambda) \right] \\ \text{s.t.} \quad & g_{\text{cone},j}(q, \lambda_j) \le 0, \quad j = 1, \cdots, n_c \end{aligned} \tag{2}$$

where $q \in \mathrm{SE}(3)$ is the candidate object's pose and $\lambda = (\lambda_1, \cdots, \lambda_{n_c}) \in \mathbb{R}^{3n_c}$ collects all contact forces in the resulting structure, including inter-object and ground contacts. This formulation follows prior contact-aware optimization methods [16, 5].

The geometric objective f_{geo} is designed as follows:

$$\begin{aligned} f_{\text{geo}}(q) = w_{\text{h}} f_{\text{h}}(q) + w_{\text{in}} f_{\text{in}}(q \mid T, O_{i_{k+1}}) \\ + w_{\text{pen}} f_{\text{pen}}(q \mid \mathcal{X}_k, O_{i_{k+1}}) \end{aligned}$$

with positive weights. For $q = (R, t)$, $f_{\text{h}}(q) = t_z$ is the world-frame height of the candidate's pose origin. Writing $O = O_{i_{k+1}}$, we penalize proximity to existing objects and the fixed ground, indexed by i_0, as

$$f_{\text{pen}}(q) = \sum_{j \in \mathcal{P}_k \cup \{i_0\}} \left[\epsilon_{\text{margin}} - \mathrm{d}(O(q), O_j(q_j)) \right]_+^2,$$

where d is signed distance (positive for separation), $[x]_+ = \max(x, 0)$, and $\epsilon_{\text{margin}} \ge 0$ is the clearance margin. With $h_{O(q)}(u) = \max_j h_{C_{i_{k+1},j}(q)}(u)$, we define

$$\begin{aligned} m_{\text{in}}(q) &= \max_j \min_{u \in \mathbb{S}^2} \left[h_{T_j}(u) - h_{O(q)}(u) \right], \\ f_{\text{in}}(q) &= \left[-m_{\text{in}}(q) \right]_+^2. \end{aligned}$$

This term penalizes violations of containment of the entire object within a single convex target component, a conservative condition for containment in T. It is differentiable almost everywhere in q; we compute pose derivatives at regular inner solutions by implicit differentiation. Nonsmooth transitions have not caused practical difficulties in our implementation.

Let $\mathcal{P}_{k+1} := \mathcal{P}_k \cup \{i_{k+1}\}$, excluding the ground index i_0. For distinct $l, m \in \mathcal{P}_{k+1} \cup \{i_0\}$, let $I_{l,m} \subset \{1, \ldots, n_c\}$ index contacts between bodies l and m. Each object's contact set $I_l := \bigcup_{m \in (\mathcal{P}_{k+1} \cup \{i_0\}) \setminus \{l\}} I_{l,m}$ includes ground contacts, but the physical objective excludes the ground's equilibrium residual:

$$\begin{aligned} f_{\text{phys}}(q, \lambda) = w_{\text{eq}} \sum_{l \in \mathcal{P}_{k+1}} \| J_{c,I_l}^{\top}(q) \lambda_{I_l} + \lambda_{\text{ext},l} \|_W^2 \\ + w_{\text{cp}} \sum_{j \in I_{i_{k+1}}} \mathrm{d}_j(q)^2 \|\lambda_j\|^2 + w_{\text{reg}} \sum_{j=1}^{n_c} \|\lambda_j\|^2 \end{aligned} \tag{3}$$

The three terms respectively penalize violations of static equilibrium, contact complementarity between the candidate and the current scene, and large contact forces through regularization. Here, $\mathrm{d}_j(q)$ is the signed distance between the two bodies associated with contact candidate j. J_{c,I_l} is the contact Jacobian for object O_l, $\lambda_{\text{ext},l}$ is its external wrench, and λ_{I_l} collects its contact forces. The weighted norm

$\|x\|_W^2 = x^\top W x$, $W \succ 0$, sets the relative force and torque penalties; $w_{\text{eq}}, w_{\text{cp}}, w_{\text{reg}} > 0$ are objective weights.

Finally, each $g_{\text{cone},j}(\cdot)$ is the smoothed friction cone constraints for the j-th contact point defined by

$$g_{\text{cone},j}(q, \lambda_j) = \sqrt{\|J_{c,j}^t \lambda_j\|^2 + \epsilon} - \mu_j J_{c,j}^n \lambda_j \tag{4}$$

where $\epsilon > 0$ is a sufficiently small smoothing parameter, μ_j is the friction coefficient, and $J_{c,j}^t$ and $J_{c,j}^n$ are the tangential and normal components of the Jacobian of the contact constraint.

B. Upper-Level Problem Optimization on SE(3) Manifold

Denoting $\lambda^*(q)$ as the optimal λ of the lower-level problem for a given pose q, we define the force dependent objective function as $f_{\text{phys}}(q) := f_{\text{phys}}(q, \lambda^*(q))$. Then, using the *Envelope theorem* [4], we can compute the gradient of $f_{\text{phys}}(q)$ with respect to q as follows:

$$\frac{df_{\text{phys}}(q)}{dq} = \frac{\partial f_{\text{phys}}(q, \lambda^*)}{\partial q} + \sum_{j=1}^{n_c} \nu_j^* \frac{\partial g_{\text{cone},j}(q, \lambda_j^*)}{\partial q}$$

where ν_j^* is the optimal dual variable associated with the j-th friction cone constraint $g_{\text{cone},j}(q, \lambda_j^*) \leq 0$. Using the above gradient, we optimize the placement pose q on SE(3) manifold based on the Riemannian trust-region method [1] combined with BFGS Hessian approximation [10]. We use a homotopy schedule to jointly adjust the geometric and physical objective weights, including the force regularization weight w_{reg}. Each stage holds these weights fixed and solves Eq. (2), initialized from the preceding stage's pose. The schedule progressively reduces w_{reg} to limit regularization-induced equilibrium residuals, and the final stage supplies the pose and contact forces used for candidate evaluation.

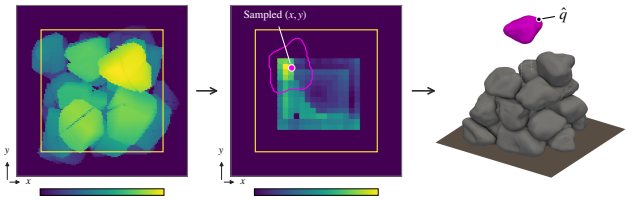
C. Lower-Level Problem Optimization with ADMM

For a fixed q, the lower-level problem is a convex optimization problem with respect to λ which can be efficiently solved. Moreover, exploiting the inherent structure of the lower-level problem, we can reformulate it to be separable with respect to the objects $\mathcal{P}_{k+1}$ in the scene and apply the alternating direction method of multipliers (ADMM) [3] to solve it in a scalable manner. For each contact force λ_j, we introduce two local copies $\lambda_{j,1}$ and $\lambda_{j,2}$ associated with the two contacting bodies, including the fixed ground. Specifically, for $j \in I_{l,m}$ with $l < m$, the copies are associated with bodies l and m, respectively. We then impose the consensus constraints $\lambda_{j,1} = \lambda_{j,2}$. Collecting all local copies into $\tilde{\lambda}$ makes the force and torque equilibrium term separable by body: each body's residual depends only on its own local contact-force variables. The lower-level problem can therefore be reformulated as

$$\begin{aligned} \min_{\tilde{\lambda}, z} \quad & \tilde{f}_{\text{phys}}(q, \tilde{\lambda}) + \sum_{\substack{j=1,\dots,n_c \\ s=1,2}} \delta_{K_j}(z_{j,s}) \\ \text{s.t.} \quad & \lambda_{j,1} = \lambda_{j,2} = z_{j,1} = z_{j,2}, \qquad j = 1, \dots, n_c. \end{aligned}$$

where $\tilde{f}_{\text{phys}}(q, \tilde{\lambda})$ is the extended physical force function and δ_{K_j} is the indicator function of the convex set of feasible contact forces $K_j := \{\lambda \in \mathbb{R}^3 \mid g_{\text{cone},j}(q, \lambda) \leq 0\}$. Here, $z_{j,s}$ is an auxiliary variable associated with the j-th contact point and $s \in \{1, 2\}$. At each ADMM iteration, the objectwise-separable subproblem decomposes into $|\mathcal{P}_{k+1}|$ independent subproblems, one for each object O_l. An equivalent four-dimensional second-order cone (SOC) lifting enables closed-form cone projections for the friction updates, while consensus updates reduce to averaging.

The pose optimization module is implemented using CUDA to leverage GPU parallelization for efficient computation of the gradients and the ADMM updates.



(a) Scene height (b) Proposal score (c) Sampled initial pose

Fig. 4: Initial pose sampling for a fixed object and orientation: (a) scene height, (b) proposal score and sampled planar location, and (c) initial pose $\hat{q}$ above the scene. Magenta links the sampled location and candidate object.

V. Sequence Planning

A. Initial Pose Sampling

The sampler π_{init} uses volume-biased object sampling with a finite orientation set per object. It prioritizes the lowest incomplete layer and seeds empty scenes near feasible target corners. For object i, orientation R, and feasible planar location $\mathbf{x}$, the score is

$$\sigma(\mathbf{x} \mid i, R) = w_{\text{s}}\sigma_{\text{s}} + w_{\text{f}}\sigma_{\text{f}} + w_{\text{b}}\sigma_{\text{b}} - w_{\text{h}}\sigma_{\text{h}} - w_{\text{d}}\sigma_{\text{d}},$$

with positive weights. The first four terms measure the fraction of bottom samples near predicted contact, unfilled layer area beneath the footprint, proximity to the feasible placement boundary, and normalized mean contact height, respectively. The distance penalty $\sigma_{\text{d}} = [d_{\text{prev}} - \delta]_+ / (L + [d_{\text{prev}} - \delta]_+)$ uses the minimum gap d_{prev} between the convex XY footprints of the candidate and preceding object, gap tolerance $\delta \geq 0$, and length scale L, the cube root of the initial inventory's mean object volume. Favoring nearby placements helps MCTS lookahead assess interactions between successive objects; distant proposals remain eligible. The first placement incurs no penalty. The floor and boundary preferences are inspired by Menezes et al. [20]. Figure 4 illustrates the proposal score and sampled initial pose.

We combine elite selection with sampling without replacement weighted by $\exp(\sigma/\tau)$, $\tau > 0$. Proposals start above the scene with clearance and undergo π_{opt} refinement, including XY/yaw mutations at the root. To preserve locality, both optimized and settled XY positions must stay within $\eta \mathbb{V}(O_i)^{1/3}$ of the proposal's initial XY position, $\eta > 0$. Refined poses' lowest points must not exceed the active-layer height plus a tolerance; projected coverage determines layer advancement.

Blocked expansion may retry without this layer restriction within the remaining budget; other admission checks remain.

B. Simulation Module

The module π_{sim} evaluates every proposed transition using the same simulation procedure at all search depths. It first settles the new object for a fixed duration while holding the existing objects static. From the resulting poses, it resets velocities and simulates the entire structure's response to the added load. Structural simulation requires a minimum duration and sustained low mass-normalized kinetic energy. Excessive accumulated motion, invalid contacts, or failure to settle within the time limit rejects the placement.

For an accepted nominal simulation, the settled scene defines the successor $s' = F(s, a)$. Each disturbance trial starts from this scene with the new object's initial world-frame linear velocity

$$\xi_j = \big(v_h \cos\theta_j,\ v_h \sin\theta_j,\ -v_d\big)^\top,$$

where $\theta_j = 2\pi j/n_d$ for $j = 0, \ldots, n_d - 1$, and $v_h, v_d > 0$ are prescribed horizontal and downward speeds. Initial angular velocity is zero, and every candidate uses the same n_d disturbances. To reduce computation, only the candidate and stones within two links in the nominal scene's contact graph of movable stones remain dynamic throughout each trial. All other bodies retain fixed collision geometry. These trials estimate local robustness without changing s'. The resulting successor, reward, and feasibility outcome are stored in the tree and used directly for subsequent search and action selection. Simulation uses CUDA subADMM [11].

C. Reward Design

For the newly placed object $i = i_{k+1}$, let q_i' be its pose in the settled successor s' returned by π_{sim} (Sec. V-B). We define the target-volume and object-volume ratios

$$\rho_T = \frac{\mathbb{V}\big(O_i(q_i') \cap T\big)}{\mathbb{V}(T)}, \quad \rho_O = \frac{\mathbb{V}\big(O_i(q_i') \cap T\big)}{\mathbb{V}(O_i)}.$$

The ρ_T term encourages target filling toward the objective in Eq. (1), while ρ_O discourages placements extending excessively outside the target. Intersection volumes are estimated on a voxel grid. With fixed normalization scales absorbed into positive weights, the standard excavator reward is

$$\begin{aligned} r(s, a) = w_T \rho_T + w_O[\rho_O - \tfrac{1}{2}]_+ - w_E e(a) \\ + w_H\big(\phi_H(s) - \phi_H(s')\big) - w_D D, \end{aligned}$$

where $[x]_+ = \max(x, 0)$, so the containment bonus is active above 50%. Coverage lost through motion of earlier objects is not subtracted. The height term rewards the decrease in surface-height variance $\phi_H(s) = \langle (H_s - \langle H_s \rangle_\Omega)^2 \rangle_\Omega$ between parent and settled successor. Here H_s is height over the target footprint Ω, zero at empty cells, and $\langle \cdot \rangle_\Omega$ denotes the area-weighted mean.

The equilibrium score uses residual forces $\mathbf{f}_l$ and torques $\boldsymbol{\tau}_l$ returned by pose optimization:

$$e(a) = \min\left\{1, \max_l \left(\frac{\|\mathbf{f}_l\|}{m_l g} + \frac{\|\boldsymbol{\tau}_l\|}{m_l g \mathbb{V}(O_l)^{1/3}}\right)\right\}.$$

Here l indexes objects in the resulting structure and g is gravitational acceleration. Let $q_l(t; \xi_j)$ denote object l's pose under the predefined velocity disturbance ξ_j (Sec. V-B), starting from its nominal successor pose q_l'. Disturbance sensitivity is

$$D = \min\left\{D_{\max}, \max_{j,l,t} d_{\text{SE(3)}}\big(q_l(t; \xi_j), q_l'\big)\right\},$$

where $d_{\text{SE(3)}}$ sums Euclidean translation distance and angular distance weighted by $\lambda_\theta > 0$. The maximum spans predefined disturbances, objects l in this local dynamic neighborhood, and simulation time; $D_{\max}$ caps the penalty. Physical rejection criteria apply independently of the reward.

D. Monte Carlo Tree Search

We use batched, receding-horizon MCTS with parallel evaluation [8] (Fig. 2).

Batched Selection. For an optimized action $a = (i, q)$, let ρ_O^{pre} denote the object-volume ratio evaluated at q, before simulation. Using the reward weights, the logit $\ell(a) = w_O \rho_O^{\text{pre}} - w_E e(a)$ defines $P(n_p, a)$ through a softmax over selectable children. Writing $N_n = N(n)$ and similarly for Q_n and $\hat{N}_n$, an evaluated child n of n_p is selected using PUCT [21]:

$$\text{PUCT}(n) = \frac{N_n Q_n}{N_n + \hat{N}_n} + c P(n_p, a) \frac{\sqrt{N_{n_p} + \hat{N}_{n_p}}}{1 + N_n + \hat{N}_n}, \quad (5)$$

where $c > 0$ controls exploration, $\hat{N}$ counts temporary reservations, and the first term is zero when its denominator is zero. At non-root nodes, progressive widening [6] increases the number of candidate actions with completed visits. Temporary reservations prevent duplicate evaluations within a batch without increasing the widening allowance.

Parallel Expansion and Evaluation. After batch selection, fresh proposals pass through $\pi_{\text{init}} \to \pi_{\text{opt}} \to \pi_{\text{sim}}$. Independent candidates are processed in parallel; selection never descends through an action without an evaluated successor. Failed candidates are excluded from further selection.

Leaf Value Estimation and Backpropagation. We replace rollouts with a discounted per-step reward heuristic, capped by depth and remaining inventory. Terminal, failed, and depth-limit leaves receive no bootstrap; exhausted shallower nonterminal leaves are not repeatedly backed up. Failures incur a discounted fixed per-step penalty over the remaining horizon, capped by inventory. Backpropagation updates Q and N and clears temporary reservations.

Action Selection and Tree Reuse. Evaluated feasible root actions are ranked by $(1 - \alpha)Q(n_a) + \alpha G_{\text{obs}}(n_a)$, $\alpha \in [0, 1]$. Here G_{obs} is the best discounted return on each candidate's deepest evaluated feasible paths within the horizon, without bootstrap; root-only paths contribute their immediate reward.

After execution, we retain successor subtrees with matching states and planning contexts. Other subtrees and branches with relaxed layer admission are discarded, resetting statistics before reuse.

Acknowledgments

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